

零声教育 Darren

课程: [埃梯零声 \(0voice.com\)](https://0voice.com).

特别注意:

课程使用的RO2版本必须在 Ubuntu24.04 desktop 桌面版运行

Ubuntu24.04桌面版下载地址: <https://mirrors.aliyun.com/ubuntu-releases/24.04/ubuntu-24.04.2-desktop-amd64.iso>

课件地址: https://gitlab.0voice.com/2503_vip/ros2-doc

源码地址: https://gitlab.0voice.com/2503_vip/ros2-src

1 安装GZ Harmonic版本

	GZ Citadel (LTS)	GZ Fortress (LTS)	GZ Garden	GZ Harmonic (LTS)
ROS 2 Jazzy (LTS)	✗	✗	⚡	✅
ROS 2 Rolling	✗	✅	⚡	⚡
ROS 2 Iron	✗	✅	⚡	⚡
ROS 2 Humble (LTS)	✗	✅	⚡	⚡
ROS 2 Foxy (LTS)	✅	✗	✗	✗
ROS 1 Noetic (LTS)	✅	⚡	✗	✗

- ✅ - Recommended combination
- ✗ - Incompatible / not possible.
- ⚡ - Possible, *but use with caution*. These combinations of ROS and Gazebo can be made to work together, but some effort is required.

```
sudo apt-get install lsb-release gnupg
```

```
sudo curl https://packages.osrfoundation.org/gazebo.gpg --output  
/usr/share/keyrings/pkgs-osrf-archive-keyring.gpg
```

```
echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/pkgs-  
osrf-archive-keyring.gpg] http://packages.osrfoundation.org/gazebo/ubuntu-stable  
$(lsb_release -cs) main" | sudo tee /etc/apt/sources.list.d/gazebo-stable.list >  
/dev/null
```

```
sudo apt-get update
```

```
sudo apt-get install gz-harmonic
```

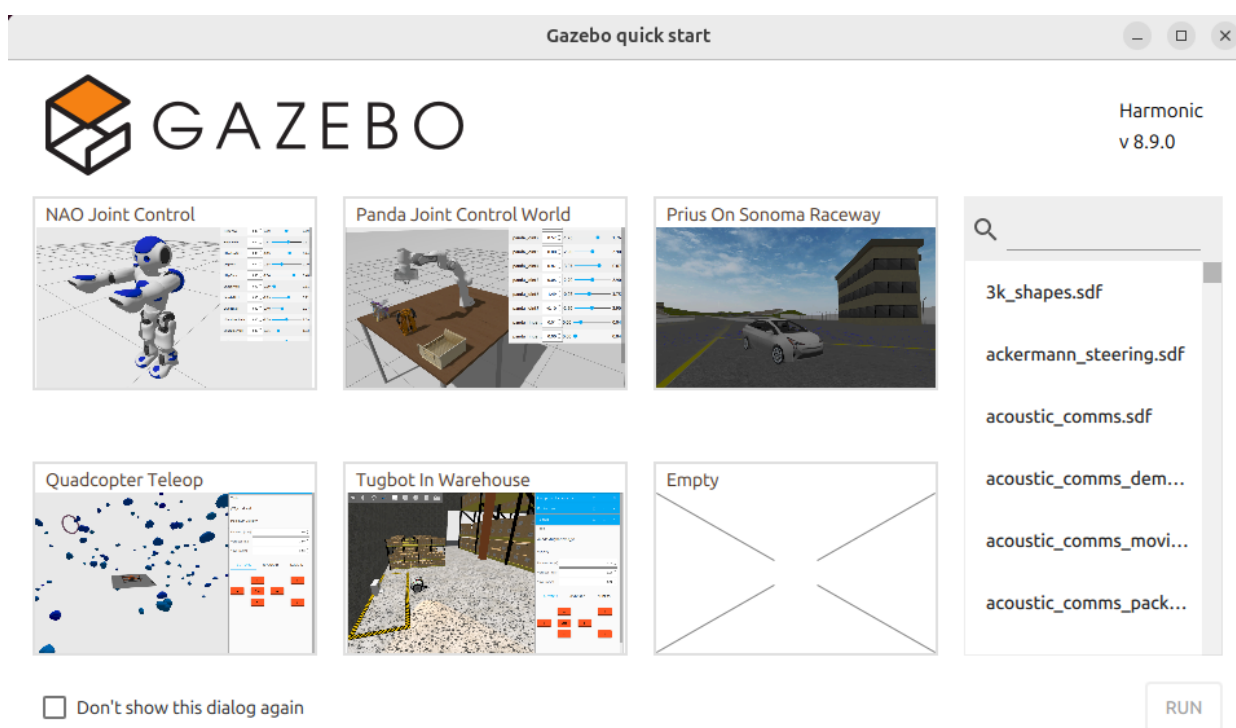
```
sudo apt-get install ros-${ROS_DISTRO}-ros-gz
```

而这也是官网最推荐的安装方式. 如果要安装其他版本, 参考[这里](#). 注意 `ros_gz` 是一个metapackage, 它安装了所有整合ROS和Gazebo的packages. 但要单独运行 `gz` 指令还需要安装:

```
sudo apt install ros-harmonic-gazebo-ros-pkgs
```

2 启动gazebo

```
lqf@lqf-ros2:~$ gz sim
```



也可以直接指定sdf (指定场景)

```
gz sim sdf shapes.sdf
```

机器人仿真示例

```
ros2 launch ros_gz_sim_demos diff_drive.launch.py
```

```
# 然后使用ros2 topic list查看当前的topic
```

```
ros2 topic list | grep cmd_vel
```

```
#得到
```

```
/model/vehicle_blue/cmd_vel
```

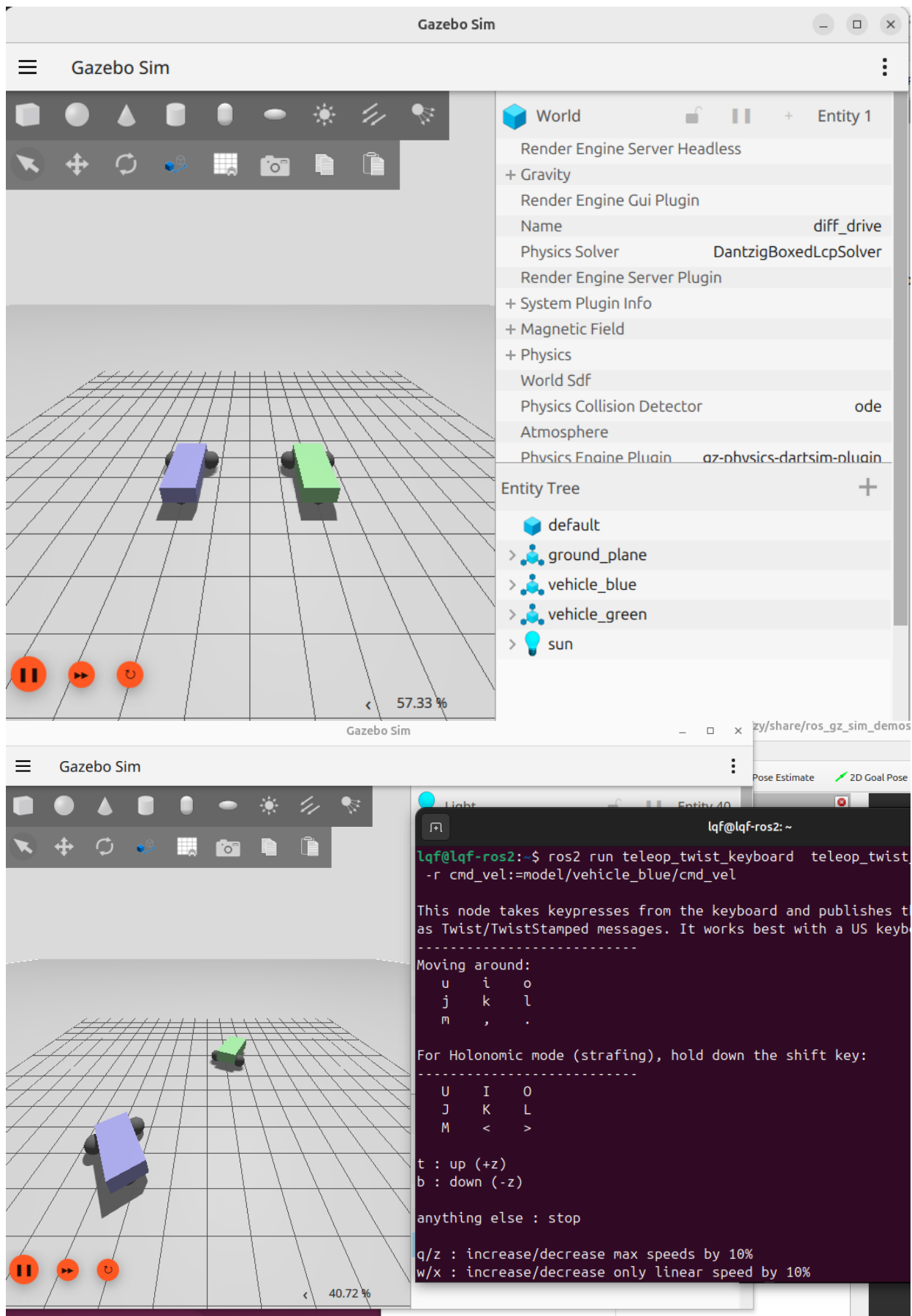
```
/model/vehicle_green/cmd_vel
```

```
# 我们先控制蓝色小车
```

```
ros2 run teleop_twist_keyboard teleop_twist_keyboard --ros-args -r  
cmd_vel:=/model/vehicle_blue/cmd_vel
```

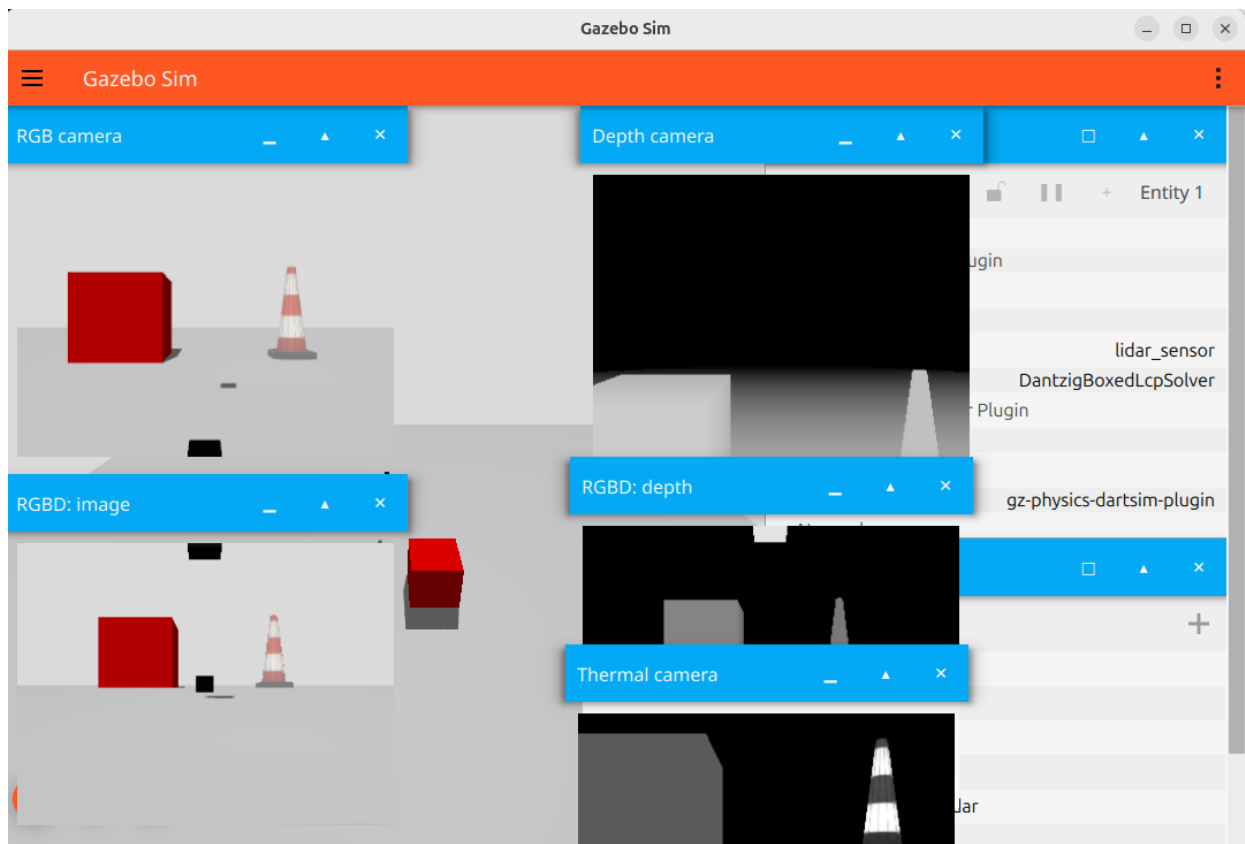
```
# 如果要控制绿色小车
```

```
ros2 run teleop_twist_keyboard teleop_twist_keyboard --ros-args -r  
cmd_vel:=/model/vehicle_green/cmd_vel
```



传感器仿真示例

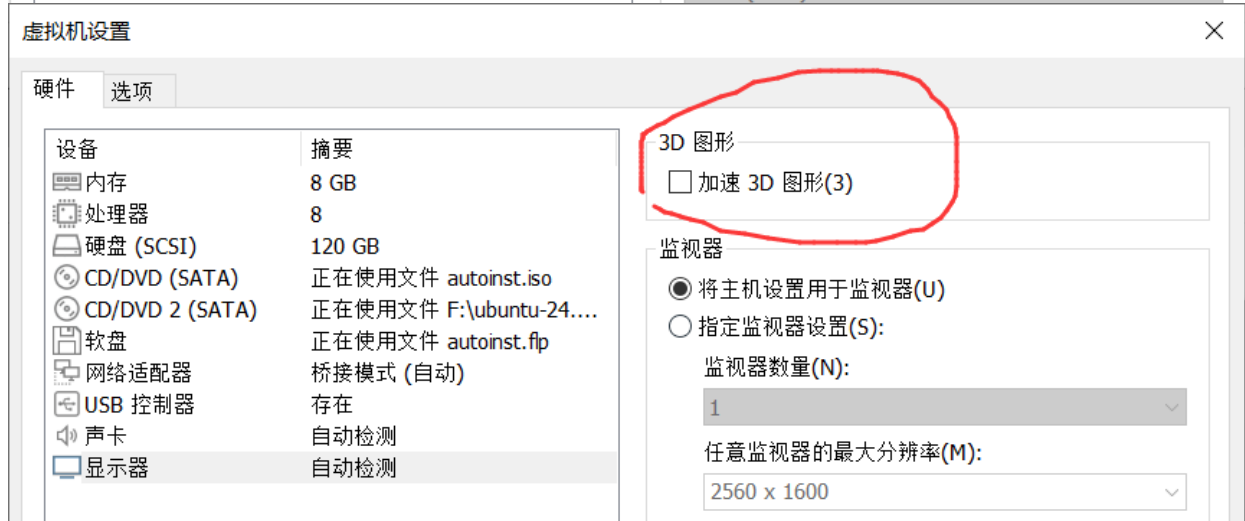
```
ros2 launch ros_gz_sim_demos rgbd_camera_bridge.launch.py
```

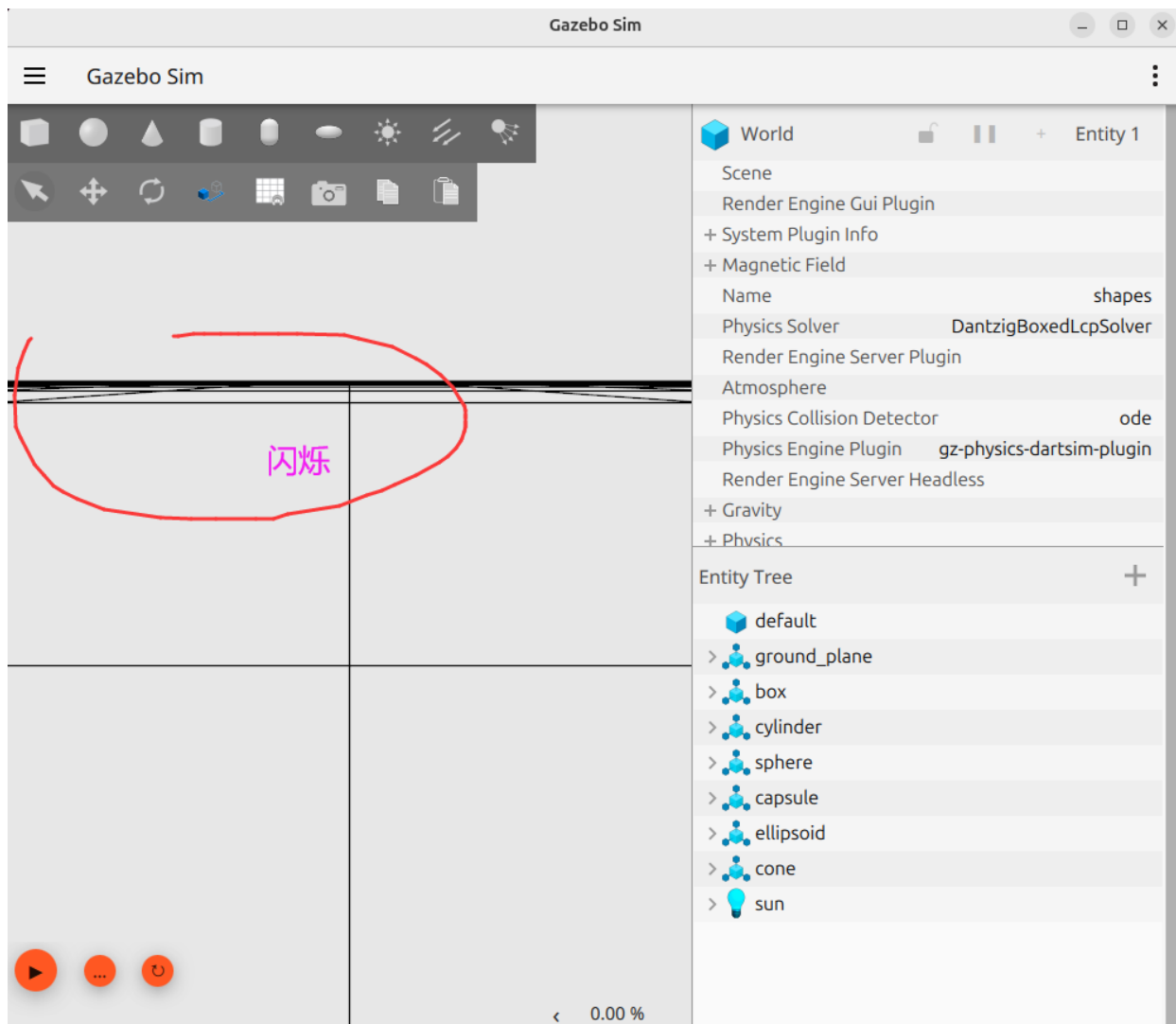


问题处理

闪烁问题

如果打开后一直闪烁，只需要在虚拟机的“显卡”设置里面，关闭“3D加速”，然后启动虚拟机，一般就可以了。





3 安装和测试TurtleBot3

3.1 安装TurtleBot3相关包

安装TurtleBot3包

```
# 安装TurtleBot3核心包
sudo apt update
sudo apt install ros-jazzy-turtlebot3*

# 安装导航相关包
sudo apt install ros-jazzy-navigation2 ros-jazzy-nav2-bringup

# 安装SLAM相关包
sudo apt install ros-jazzy-slam-toolbox

# 安装Dynamixel SDK (TurtleBot3电机控制)
sudo apt install ros-jazzy-dynamixel-sdk
```

3.2 设置环境变量

设置TurtleBot3模型

```
# 设置TurtleBot3模型
echo 'export TURTLEBOT3_MODEL=waffle' >> ~/.bashrc
echo 'export ROS_DOMAIN_ID=30 #TURTLEBOT3' >> ~/.bashrc
echo 'export
GAZEBO_MODEL_PATH=$GAZEBO_MODEL_PATH:/opt/ros/jazzy/share/turtlebot3_gazebo/models'
>> ~/.bashrc
# 重新加载配置
source ~/.bashrc

# 验证环境变量
echo $TURTLEBOT3_MODEL
```

可选的TurtleBot3模型：

- `burger`：最小型号
- `waffle`：标准型号（推荐）
- `waffle_pi`：带激光雷达的型号

3.4 测试基础仿真

启动Gazebo仿真

```
# 启动
ros2 launch turtlebot3_gazebo turtlebot3_world.launch.py
```

启动遥控节点

```
# 新开终端，启动键盘控制
ros2 run turtlebot3_teleop teleop_keyboard
```

3.5 SLAM建图演示

启动SLAM

```
# 终端1: 启动Gazebo仿真
ros2 launch turtlebot3_gazebo turtlebot3_world.launch.py

# 终端2: 启动SLAM 启动RViz可视化
ros2 launch turtlebot3_cartographer cartographer.launch.py use_sim_time:=True

# 终端3: 键盘控制机器人移动建图
ros2 run turtlebot3_teleop teleop_keyboard
```

保存地图

```
# 建图完成后保存地图
ros2 run nav2_map_server map_saver_cli -f ~/my_map
```

3.6 导航演示

使用保存的地图进行导航

```
# 终端1: 启动Gazebo仿真
ros2 launch turtlebot3_gazebo turtlebot3_world.launch.py

# 终端2: 启动导航
ros2 launch turtlebot3_navigation2 navigation2.launch.py use_sim_time:=True
map:=$HOME/my_map.yaml
```

7. 常用launch文件

创建自定义launch文件

```
# ~/turtlebot3_demo.launch.py
from launch import LaunchDescription
from launch.actions import IncludeLaunchDescription
from launch.launch_description_sources import PythonLaunchDescriptionSource
from launch_ros.substitutions import FindPackageShare
import os

def generate_launch_description():
    # 获取包路径
    turtlebot3_gazebo_path = FindPackageShare('turtlebot3_gazebo')
    turtlebot3_navigation2_path = FindPackageShare('turtlebot3_navigation2')

    return LaunchDescription([
        # 启动Gazebo仿真
        IncludeLaunchDescription(
            PythonLaunchDescriptionSource([
                turtlebot3_gazebo_path, '/launch/turtlebot3_world.launch.py'
            ])
        )
    ])
```

```
),

# 启动导航
IncludeLaunchDescription(
  PythonLaunchDescriptionSource([
    turtlebot3_navigation2_path, '/launch/navigation2.launch.py'
  ]),
  launch_arguments={
    'use_sim_time': 'True',
    'map': os.path.expanduser('~/.my_map.yaml')
  }.items()
)
])
```

3.8 故障排除

检查安装状态

```
# 检查TurtleBot3包
ros2 pkg list | grep turtlebot3

# 检查环境变量
echo $TURTLEBOT3_MODEL
echo $GAZEBO_MODEL_PATH

# 检查话题
ros2 topic list

# 检查节点
ros2 node list
```

常见问题解决

1. Gazebo启动失败

```
# 检查Gazebo版本
gazebo --version

# 重置Gazebo
killall gzserver gzclient
```

2. 模型加载失败

```
# 手动设置模型路径
export
GAZEBO_MODEL_PATH=$GAZEBO_MODEL_PATH:/opt/ros/jazzy/share/turtlebot3_gazebo/models

# 或者重新安装模型
sudo apt reinstall ros-jazzy-turtlebot3-gazebo
```

3. 找不到launch文件

```
# 查找launch文件位置
find /opt/ros/jazzy -name "*.launch.py" | grep turtlebot3

# 或者使用ros2命令查找
ros2 pkg prefix turtlebot3_gazebo
```

3.9 验证安装完成

完整测试流程

```
# 1. 启动仿真
ros2 launch turtlebot3_gazebo turtlebot3_world.launch.py

# 2. 新终端测试遥控
ros2 run turtlebot3_teleop teleop_keyboard

# 3. 新终端查看传感器数据
ros2 topic echo /scan

# 4. 查看摄像头数据（如果有）
ros2 topic echo /camera/image_raw
```

完成以上步骤后，您就可以在Ubuntu 24.04 + ROS2 Jazzy环境下使用TurtleBot3进行仿真、SLAM建图和自主导航了。

3.10 额外功能

多机器人仿真

```
# 可以尝试多机器人仿真
ros2 launch turtlebot3_gazebo multi_turtlebot3.launch.py
```

4. 创建工作空间并获取源码（可选）

暂时不理睬这里的文档

如果包安装有问题，可以从源码编译：

```
# 创建工作空间
mkdir -p ~/turtlebot3_ws/src
cd ~/turtlebot3_ws/src

# 克隆TurtleBot3仿真源码
git clone -b jazzy https://github.com/ROBOTIS-GIT/turtlebot3_simulations.git

# 克隆TurtleBot3核心包
```

```
git clone -b jazzy https://github.com/ROBOTIS-GIT/turtlebot3.git
```

```
cd ~/turtlebot3_ws
```

```
# 初始化rosdep（需要sudo权限）
```

```
sudo rosdep init
```

```
# 更新rosdep数据库，这里需要手动更新了
```

```
rosdep update
```

```
# 安装依赖
```

```
rosdep install --from-paths src --ignore-src -r -y
```

```
# 编译
```

```
colcon build --symlink-install
```

```
# 设置环境
```

```
echo 'source ~/turtlebot3_ws/install/setup.bash' >> ~/.bashrc
```

```
source ~/.bashrc
```

参考文档

[ROS Package: ros_gz_sim demos](#)